

Homework 1

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Problem 1 (Examples Of Metric Spaces). Prove that the following are metric spaces

- (i) $(\mathbb{R}^n, d_\infty)$ where $d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$ for all $x, y \in \mathbb{R}^n$.

Proof. Notice that for all $x, y \in \mathbb{R}^n$, $d_\infty(x, y) \geq 0$ (that is, $\max_{1 \leq i \leq n} |x_i - y_i| \geq 0$) because $|x_i - y_i| \geq 0$ for all $1 \leq i \leq n$.

- (I) Let $x, y \in \mathbb{R}^n$ be such that $d_\infty(x, y) = 0$. Then this holds if and only if

$$\begin{aligned} \max_{1 \leq i \leq n} |x_i - y_i| = 0 &\iff |x_i - y_i| = 0 \quad \forall 1 \leq i \leq n \\ &\iff x_i = y_i \quad \forall 1 \leq i \leq n \\ &\iff x = y. \end{aligned}$$

- (II) Observe that for all $x, y \in \mathbb{R}^n$, we have

$$d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i| = \max_{1 \leq i \leq n} |y_i - x_i| = d_\infty(y, x).$$

- (III) Let $x, y, z \in \mathbb{R}^n$. Observe that for all $1 \leq i \leq n$

$$\begin{aligned} |x_i - z_i| &\leq |x_i - y_i| + |y_i - z_i| \\ &\leq \max_{1 \leq i \leq n} |x_i - y_i| + \max_{1 \leq i \leq n} |y_i - z_i|. \end{aligned}$$

Thus,

$$|x_i - z_i| \leq \max_{1 \leq i \leq n} |x_i - y_i| + \max_{1 \leq i \leq n} |y_i - z_i|. \quad (1)$$

Notice that the right-hand side above is an upper bound for the set

$$\{1 \leq i \leq n : |x_i - z_i|\}.$$

Thus, (1) implies that

$$\max_{1 \leq i \leq n} |x_i - z_i| \leq \max_{1 \leq i \leq n} |x_i - y_i| + \max_{1 \leq i \leq n} |y_i - z_i|.$$

By definition of d_∞ , we obtain

$$d_\infty(x, z) \leq d_\infty(x, y) + d_\infty(y, z).$$

From (I) - (III), we can conclude that $(\mathbb{R}^n, d_\infty)$ is a metric space. ■

- (ii) $(C([a, b]), d_\infty)$ where $d_\infty(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)|$ for all $f, g \in C([a, b])$.

Proof. Notice that for all $t \in [a, b]$, $|f(t) - g(t)| \geq 0$ and so

$$\sup_{t \in [a, b]} |f(t) - g(t)| \geq 0 \implies d_\infty(f, g) \geq 0.$$

(I) Let $f, g \in C([a, b])$ be such that $d_\infty(f, g) = 0$. This holds if and only if

$$\begin{aligned} \sup_{t \in [a, b]} |f(t) - g(t)| = 0 &\iff |f(t) - g(t)| = 0 \quad \forall t \in [a, b] \\ &\iff f(t) = g(t) \quad \forall t \in [a, b] \\ &\iff f = g. \end{aligned}$$

(II) Observe that for all $f, g \in C([a, b])$, we have

$$d_\infty(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)| = \sup_{t \in [a, b]} |g(t) - f(t)| = d_\infty(g, f).$$

(III) Let $f, g, h \in C([a, b])$. Using the triangle inequality on the standard metric $|\cdot|$ of $\mathbb{R}$, we get that for all $t \in [a, b]$

$$\begin{aligned} |f(t) - g(t)| &\leq |f(t) - h(t)| + |h(t) - g(t)| \\ &\leq \sup_{t \in [a, b]} |f(t) - h(t)| + \sup_{t \in [a, b]} |h(t) - g(t)| \\ &= d_\infty(f, h) + d_\infty(h, g). \end{aligned}$$

Thus, we get

$$|f(t) - g(t)| \leq d_\infty(f, h) + d_\infty(h, g).$$

Since the right-hand side of the above inequality is an upper bound for the set

$$\{t \in [a, b] : |f(t) - g(t)|\}$$

we get that

$$\sup_{t \in [a, b]} |f(t) - g(t)| \leq d_\infty(f, h) + d_\infty(h, g);$$

that is,

$$d_\infty(f, g) \leq d_\infty(f, h) + d_\infty(h, g).$$

From (I) - (III), we can conclude that $(C([a, b]), d_\infty)$ is a metric space. ■

Problem 2 (Examples of Open Sets in Metric Spaces). (i) Let $X \neq \emptyset$ and d be the discrete metric on X . Prove that any subset of X is open.

Proof. Let A be any subset of X . Our goal is to show that for all $a \in A$, there exists an $r_a > 0$ such that $B_{r_a}(a) \subseteq A$. To this end, let $a \in A$. Define $r = 1$ which is clearly positive. Now, we claim that $B_r(a) \subseteq A$. Let $x \in B_r(a)$ be arbitrary. By definition of the discrete metric, if $d(a, x) = 1$, then we have $1 < 1$ which is a contradiction. The only other case for which $d(x, a) < 1$ is when $d(a, x) = 0$. But this tells us that $x = a$ and so $x \in A$. Hence, $B_r(a) \subseteq A$ and so A must be an open set. ■

(ii) Prove that $\{(x_1, x_2) \in \mathbb{R}^2 : 0 < x_1 < 1\}$ is open in $(\mathbb{R}^2, d_\infty)$ where $d_\infty(x, y) = \max_{i=1,2} |x_i - y_i|$ for all $x, y \in \mathbb{R}^2$.

Proof. Set $A = \{(x_1, x_2) \in \mathbb{R}^2 : 0 < x_1 < 1\}$. To show that A is open, we need to show that for all $(x_1, x_2) \in A$, there exists $\delta > 0$ such that $B_\delta((x_1, x_2)) \subseteq A$. Let $(x_1, x_2) \in A$. Define $\delta = \min\{1 - x_1, x_1\}$. Since $0 < x_1 < 1$, we know that $1 - x_1 > 0$ and $x_1 > 0$ and so $\delta > 0$. From here, we claim that $B_\delta((x_1, x_2)) \subseteq A$. To this end, let $(y_1, y_2) \in B_\delta((x_1, x_2))$. Then we have

$$d_\infty((x_1, x_2), (y_1, y_2)) < \delta \iff \max_{i=1,2} |x_i - y_i| < \delta.$$

This tells us that

$$|x_1 - y_1| \leq \max_{i=1,2} |x_i - y_i| < \delta \implies |x_1 - y_1| < \delta.$$

Then we get

$$x_1 - \delta < y_1 < x_1 + \delta.$$

Since $\delta < x_1$ and $\delta < 1 - x_1$, we get that

$$0 = x_1 - x_1 < x_1 - \delta < y_1 < x_1 + \delta < x_1 + (1 - x_1) = 1.$$

Thus, we have $0 < y_1 < 1$ and so $(y_1, y_2) \in A$. Therefore, $B_\delta((x_1, x_2)) \subseteq A$ and so A is open. ■

- (iii) Prove that $\{(x_1, x_2) \in \mathbb{R}^2 : x_2 > 0\}$ is open in $(\mathbb{R}^2, d_1)$ where $d_1(x, y) = \sum_{i=1}^2 |x_i - y_i|$ for all $x, y \in \mathbb{R}^2$.

Proof. Set $A = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > 0\}$. Our goal is to show that for all $(x_1, x_2) \in A$, there exists a $\delta > 0$ such that $B_\delta((x_1, x_2)) \subseteq A$. Define $\delta = x_2$. To this end, let $(y_1, y_2) \in B_\delta((x_1, x_2))$. To show that $(y_1, y_2) \in A$, we need to show that $y_2 > 0$. We get that

$$\begin{aligned} (y_1, y_2) \in B_\delta((x_1, x_2)) &\iff d_1((x_1, x_2), (y_1, y_2)) < \delta \\ &\iff \sum_{i=1}^2 |x_i - y_i| < \delta \\ &\iff |x_1 - y_1| + |x_2 - y_2| < \delta. \end{aligned}$$

Since $|x_2 - y_2| \leq |x_1 - y_1| + |x_2 - y_2|$, we have $|x_2 - y_2| < \delta$ and so

$$\begin{aligned} x_2 - \delta < y_2 < x_2 + \delta &\implies y_2 > x_2 - \delta = x_2 - x_2 = 0. \\ &\implies y_2 > 0. \end{aligned}$$

Hence, we get that $(y_1, y_2) \in A$ and so A is open. ■

Problem 3. Determine whether the following sets are open and prove your claims.

- (i) $\{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq 1\}$ in $(\mathbb{R}^2, d_\infty)$ where $d_\infty(x, y) = \max_{i=1,2} |x_i - y_i|$ for all $x, y \in \mathbb{R}^2$.

Proof. We claim that the set above is not open. Consider the point $(0, 1)$. Note that for all $r > 0$, $1 + r > 1$. Hence, $B_r((0, 1)) \not\subseteq \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq 1\}$. Thus, $\{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq 1\}$ is not an open set in $\mathbb{R}^2$. ■

- (ii) $\bigcap_{n \in \mathbb{N}} \left(\frac{1}{n^2}, 2\right)$ in $\mathbb{R}$ equipped with the Euclidean metric.

Proof. We claim that the intersection above is open. That is, we claim that

$$\bigcap_{n \in \mathbb{N}} \left(\frac{1}{n^2}, 2\right) = (1, 2)$$

where $(1, 2)$ is an open set. We proceed via double inclusion.

Let $x \in \bigcap_{n \in \mathbb{N}} \left(\frac{1}{n^2}, 2\right)$. Then for all $n \in \mathbb{N}$, $x \in \left(\frac{1}{n^2}, 2\right)$. In particular, for $n = 1$, $x \in (1, 2)$. Hence, we conclude that

$$\bigcap_{n \in \mathbb{N}} \left(\frac{1}{n^2}, 2\right) \subseteq (1, 2). \quad (1)$$

Let $x \in (1, 2)$. Since for all $n \in \mathbb{N}$ $\frac{1}{n^2} < 1$, we obtain

$$\frac{1}{n^2} < 1 < x < 2$$

which implies that $x \in \left(\frac{1}{n^2}, 2\right)$. Thus, we get

$$(1, 2) \subseteq \bigcap_{n \in \mathbb{N}} \left(\frac{1}{n^2}, 2\right). \quad (2)$$

Now, (1) and (2) imply that $\bigcap_{n \in \mathbb{N}} \left(\frac{1}{n^2}, 2\right)$ is indeed an open set in $\mathbb{R}$. ■

Problem 4 (In a Metric Space, open balls are open indeed). Let (X, d) be a metric space, $x_0 \in X$, $r > 0$. Prove that $B_r(x_0)$ is open.

Proof. Our goal is to show that for all $x \in B_r(x_0)$, there exists a $r_x > 0$ such that $B_{r_x}(x) \subseteq B_r(x_0)$. To this end, let $x \in B_r(x_0)$ be arbitrary. Consider the radius $r_x = r - d(x, x_0)$. Since x lies inside $B_r(x_0)$, it follows that $d(x, x_0) < r$. Otherwise, $r_x < 0$ which will be a contradiction. Thus, r_x must be strictly positive.

From here, we claim that $B_{r_x}(x) \subseteq B_r(x_0)$. To this end, let $y \in B_{r_x}(x)$ be arbitrary. Our goal is to show that $d(y, x_0) < r$ in order for $y \in B_r(x_0)$. By the triangle inequality and the definition of r_x , we obtain

$$\begin{aligned} d(y, x_0) &\leq d(y, x) + d(x, x_0) \\ &< r_x + d(x, x_0) \\ &= (r - d(x, x_0)) + d(x, x_0) \\ &= r. \end{aligned}$$

Indeed, we get that $y \in B_r(x_0)$ and so we conclude that $B_r(x_0)$ is an open set. ■

Problem 5 (Equivalent Metrics). Let $X \neq \emptyset$. We say that two metrics d_1 and d_2 on X are **equivalent** if there exist constants $c, C > 0$ such that

$$cd_2(x, y) \leq d_1(x, y) \leq Cd_2(x, y)$$

for all $x, y \in X$.

- (i) Assume that d_1 and d_2 are equivalent metrics on $X \neq \emptyset$ and let $A \subseteq X$. Prove that A is open in (X, d_2) if and only if A is open in (X, d_1) .

Proof. ($\Leftarrow$) Suppose A is open in (X, d_1) . Our goal is to show that A is an open set in (X, d_2) . To this end, we need to show that for all $x \in A$, there exists a $\delta_x > 0$ such that $B_{\delta_x}(x) \subseteq A$. Let $x \in A$ be arbitrary. Since A is open in (X, d_1) , it follows that there exists an $\hat{\delta}_x > 0$ such that $B_{\hat{\delta}_x}(x) \subseteq A$ with respect to d_1 . Since d_1 and d_2 are equivalent metrics, there exists constants $c, C > 0$ such that

$$cd_2(x, y) \leq d_1(x, y) \leq Cd_2(x, y). \quad (*)$$

Define $\delta_x = \frac{\hat{\delta}_x}{C}$. Notice that $\delta_x > 0$ because $\hat{\delta}_x > 0$ and $C > 0$. To show that $B_{\delta_x}(x) \subseteq A$, it suffice to show that

$$B_{\delta_x}(x) \subseteq B_{\hat{\delta}_x}(x).$$

To this end, let $y \in B_{\delta_x}(x)$ be arbitrary. Then using our definition of δ_x and (*), it follows that

$$d_1(x, y) \leq Cd_2(x, y) < C \cdot \frac{\hat{\delta}_x}{C} = \hat{\delta}_x.$$

Hence, we conclude that $y \in B_{\hat{\delta}_x}(x)$ and so we conclude that A is open in (X, d_2) . ($\implies$) Suppose A is open in (X, d_2) . Let $x \in A$. Our goal is to show that there exists a $\delta_x > 0$ such that $B_{\delta_x}(x) \subseteq A$ with respect to d_1 . Since A is open in (X, d_2) , there exists a $\hat{\delta}_x > 0$ such that $B_{\hat{\delta}_x}(x) \subseteq A$ with respect to d_2 . Define $\delta_x = c\hat{\delta}_x$. We know that this is strictly positive because $\hat{\delta}_x > 0$ and $c > 0$. To show our result, it suffices to show that

$$B_{\delta_x}(x) \subseteq B_{\hat{\delta}_x}(x).$$

Indeed, if let $y \in B_{\delta_x}(x)$ be arbitrary, then using (*), it follows that

$$d_2(x, y) \leq \frac{1}{c}d_1(x, y) < \frac{1}{c} \cdot \delta_x = \frac{1}{c} \cdot c\hat{\delta}_x = \hat{\delta}_x.$$

Then $y \in B_{\hat{\delta}_x}(x)$ and so A is open in d_1 . ■

- (ii) Let d be a metric on $X \neq \emptyset$ and for $\lambda > 0$ define the function $d_\lambda(x, y) = \lambda d(x, y)$ for all $x, y \in X$. Prove that d_λ is a metric on X that is equivalent to d .

Proof. First, we show that d_λ is also a metric on $X \neq \emptyset$. Notice that for all $x, y \in X$, $d_\lambda(x, y) \geq 0$ because $\lambda > 0$ and $d(x, y) \geq 0$.

- (i) Let $x, y \in X$ be such that $d_\lambda(x, y) = 0$. Using the definition of d_λ and the nondegeneracy of d , we obtain

$$\begin{aligned} d_\lambda(x, y) = 0 &\iff \lambda d(x, y) = 0 \\ &\iff d(x, y) = 0 \\ &\iff x = y. \end{aligned}$$

- (ii) For all $x, y \in X$, we can use the symmetric property of d to get

$$d_\lambda(x, y) = \lambda d(x, y) = \lambda d(y, x) = d_\lambda(y, x).$$

- (iii) Let $x, y, z \in X$. Using the triangle inequality of d , we get that

$$\begin{aligned} d_\lambda(x, z) &= \lambda d(x, z) \leq \lambda[d(x, y) + d(y, z)] \\ &= \lambda d(x, y) + \lambda d(y, z) \\ &= d_\lambda(x, y) + d_\lambda(y, z). \end{aligned}$$

Hence, we conclude that d_λ is indeed a metric on X .

For d_λ and d being equivalent, notice that if $\lambda > 1$, then

$$\frac{1}{\lambda^2} \cdot d_\lambda(x, y) = \frac{1}{\lambda^2} \cdot \lambda d(x, y) = \frac{1}{\lambda} \cdot d(x, y) \leq d(x, y) \leq \lambda d(x, y) = d_\lambda(x, y).$$

Thus, we get that

$$\frac{1}{\lambda^2} \cdot d_\lambda(x, y) \leq d(x, y) \leq 1 \cdot d_\lambda(x, y).$$

where $c_1 = \frac{1}{\lambda^2}$ and $c_2 = 1$.

On the other hand, if $0 < \lambda < 1$, then we have

$$d_\lambda(x, y) = \lambda d(x, y) \leq d(x, y) = \frac{\lambda}{\lambda} d(x, y) \leq \frac{1}{\lambda^2} \lambda d(x, y) = \frac{1}{\lambda^2} d_\lambda(x, y)$$

with $c_1 = 1$ and $c_2 = \frac{1}{\lambda^2}$. Hence, we find that d_λ and d are equivalent metrics. ■